

Lecture 7: From Static to Dynamic Optimization

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- ▶ Brief introduction to finite-horizon dynamic optimization in discrete time under certainty
- ▶ Introduce the Bellman equation and the Principle of Optimality

- 1 Problem Description
- 2 Recursive Formulation and Dynamic Programming
- 3 Alternative Formulation
- 4 Examples

A finite-horizon dynamic optimization problem

$$\begin{aligned} \sup_{\{x_t\}_{t=1}^T} \quad & \sum_{t=0}^{T-1} F_t(x_t, x_{t+1}) \\ \text{s.t.} \quad & x_{t+1} \in \Gamma_t(x_t) \text{ for all } t = 0, 1, \dots, T-1 \\ & x_0 \text{ given} \end{aligned}$$

- ▶ $x_t \in X$ is the state variable (state vector)
- ▶ $F_t : X \times X \rightarrow \mathbb{R}$ is the instantaneous payoff function
- ▶ $\Gamma_t(x)$ is a non-empty set-valued mapping (correspondence): $\Gamma_t : X \rightrightarrows X$
- ▶ The constraint $x_{t+1} \in \Gamma_t(x_t)$ captures feasibility: tomorrow's state must be reachable from today's state
- ▶ We use sup instead of max because the maximum may not be attained in general settings

Example: Cake Eating Problem

- ▶ Finite time horizon: $t = 0, 1, \dots, T$
- ▶ At $t = 0$ the agent is given a complete cake with size $\bar{x}$
- ▶ Let x_t denote the size of the cake at the beginning of each period, so that, in particular, $x_0 = \bar{x}$
- ▶ After choosing to consume c_t of the cake in period t there is

$$x_{t+1} = x_t - c_t$$

left in period $t + 1$

- ▶ Consuming quantity c of the cake gives current utility $u(c)$

- ▶ The agent's problem can be written as

$$\max_{\{c_t\}} \sum_{t=0}^{T-1} \beta^t u(c_t)$$

subject to

$$x_{t+1} = x_t - c_t \quad 0 \leq c_t \leq x_t, \quad x_0 = \bar{x}$$

- ▶ Formulate this in the general framework:
 - ▶ $F_t(x_t, x_{t+1}) = \beta^t u(x_t - x_{t+1})$
 - ▶ $\Gamma_t(x_t) = [0, x_t]$
 - ▶ We write $c_t = x_t - x_{t+1}$ in terms of the states
 - ▶ The feasible set $\Gamma_t(x_t) = [0, x_t]$ encodes both non-negativity and resource constraints

The key trade-off in the cake-eating problem is:

- ▶ Delaying consumption is costly because of the discount factor
- ▶ But delaying some consumption is also attractive because u is concave. The concavity of u implies that the consumer gains value from consumption smoothing, which means spreading consumption out over time
- ▶ The optimal solution balances these opposing forces

- ▶ In this example, the constraint $x_{t+1} \in \Gamma_t(x_t)$ can be written as $x_{t+1} \geq 0$ and $x_{t+1} \leq x_t$
- ▶ The Lagrangian $L(x_1, x_2, \dots, x_T, \lambda_1, \dots, \lambda_T, \mu_1, \dots, \mu_T)$
- ▶ We need to solve for $3T$ unknowns from the first order conditions: T state variables and $2T$ multipliers
- ▶ As T grows, this becomes a large simultaneous system whose dimension grows with T
- ▶ Dynamic programming offers a more efficient approach by breaking the problem into stages

- ▶ Consider an example where $T = 3$
- ▶ Adopt the CRRA utility function:

$$u(c) = \frac{c^{1-\gamma}}{1-\gamma} \quad (0 < \gamma < 1)$$

- ▶ CRRA = Constant Relative Risk Aversion; γ measures the degree of risk aversion
- ▶ Higher γ means stronger preference for consumption smoothing
- ▶ The case $\gamma = 1$ corresponds to $u(c) = \ln(c)$ (logarithmic utility)
- ▶ Formulate the Lagrangian

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- ▶ Given the recursive structure of this problem, we introduce the **value function** $V_t(x)$:

$$\begin{aligned} V_t(x) &:= \sup_{\{x_s\}_{s=t+1}^T} \sum_{s=t}^{T-1} F_s(x_s, x_{s+1}) \\ \text{s.t. } &x_{s+1} \in \Gamma_s(x_s) \text{ for all } s = t, \dots, T-1 \\ &x_t = x \end{aligned}$$

- ▶ $V_t(x)$ gives the lifetime value at time t if the state is $x_t = x$: “what is the best I can do from time t onward, starting with state x ?”
- ▶ This is the *value of the problem* viewed from period t with initial condition $x_t = x$
- ▶ Note that $V_0(x_0)$ is the value of the original problem

Dynamic Programming Principle

If $(x_1, x_2, \dots, x_T)$ is a solution to problem $V_0(x_0)$, then for any $t = 1, 2, \dots, T - 1$, $(x_{t+1}, \dots, x_T)$ is a solution to problem $V_t(x_t)$.

Intuition: an optimal path remains optimal from any point along it.

Proof by contradiction:

- ▶ Suppose that there is an admissible sequence $(z_{t+1}, \dots, z_T)$ that achieves higher value than $(x_{t+1}, \dots, x_T)$ at time t when the state is x_t
- ▶ Show that $(x_0, x_1, \dots, x_t, z_{t+1}, \dots, z_T)$ is admissible and achieves higher value of $V_0(x_0)$
- ▶ In other words, we could "splice" this better tail onto our path, improving overall value, contradicting optimality

Bellman Equation

For all $t = 0, 1, \dots, T - 1$, we have

$$V_t(x) = \sup_{x' \in \Gamma_t(x)} \{F_t(x, x') + V_{t+1}(x')\},$$

where we use the terminal convention $V_T(x) = 0$.

Interpretation: today's value = today's payoff + future value.

- ▶ Choose the next period state x' to maximize: current return $F_t(x, x')$ plus continuation value $V_{t+1}(x')$
- ▶ This reduces a T -period problem to a sequence of 2-period problems
- ▶ $V_{t+1}(x')$ summarizes all future consequences of choosing x' today
- ▶ Note that we do not assume the existence of an optimal sequence here

Fix $t = 0, 1, \dots, T - 1$ and a state x .

► First, we prove that $V_t(x) \geq \sup_{x' \in \Gamma_t(x)} \{F_t(x, x') + V_{t+1}(x')\}$

① For any admissible sequence $(x, x_{t+1}, \dots, x_T)$, show that

$$V_t(x) \geq F_t(x, x_{t+1}) + \sum_{s=t+1}^{T-1} F_s(x_s, x_{s+1})$$

② Taking the supremum over $(x_{t+2}, \dots, x_T)$ gives $V_t(x) \geq F_t(x, x_{t+1}) + V_{t+1}(x_{t+1})$

③ Taking the supremum again over x_{t+1} proves the result

► Next, we prove that $V_t(x) \leq \sup_{x' \in \Gamma_t(x)} \{F_t(x, x') + V_{t+1}(x')\}$

① For any $\epsilon > 0$, let $(x, x_{t+1}, \dots, x_T)$ be an admissible sequence such that

$$V_t(x) \leq F_t(x, x_{t+1}) + \sum_{s=t+1}^{T-1} F_s(x_s, x_{s+1}) + \epsilon$$

② Show that $\sum_{s=t+1}^{T-1} F_s(x_s, x_{s+1}) \leq V_{t+1}(x_{t+1})$

③ Taking the supremum over x_{t+1} and taking the limit as $\epsilon \rightarrow 0$ completes the proof

- ▶ Using the Bellman equation, we can compute the value function through backward induction.
- ▶ In the last period, $V_{T-1}(x) = \sup_{x' \in \Gamma_{T-1}(x)} F_{T-1}(x, x')$. This is a static optimization problem. Solving this problem gives: (1) the value at $T - 1$ as a function of x_{T-1} , and (2) the optimal x_T as a function of x_{T-1} .
- ▶ Then, for each $t = 0, 1, \dots, T - 2$, we can solve for $V_t(x)$ and obtain the optimal x_{t+1} as a function of x_t :

$$V_t(x) = \sup_{x' \in \Gamma_t(x)} \{F_t(x, x') + V_{t+1}(x')\}$$

- ▶ Given x_0 , we can also obtain an admissible sequence by solving these Bellman equations.
- ▶ The question is: is this sequence our solution?

Principle of Optimality

The sequence $(x_1, \dots, x_T)$ is a solution to $V_0(x_0)$ if and only if for all $t = 0, 1, \dots, T - 1$, x_{t+1} is a solution to

$$\sup_{x' \in \Gamma_t(x_t)} \{F_t(x_t, x') + V_{t+1}(x')\},$$

where $V_T = 0$.

This is the characterization theorem: it tells us exactly when a sequence is optimal

Proof:

- ▶ Let $(x_1, \dots, x_T)$ be a solution to $V_0(x_0)$. The dynamic programming principle implies that $V_t(x_t) = \sum_{s=t}^{T-1} F_s(x_s, x_{s+1})$ for all $t = 0, 1, \dots, T-1$. By definition of the value function and Bellman equation:

$$F_t(x_t, x_{t+1}) + V_{t+1}(x_{t+1}) = V_t(x_t) = \sup_{x' \in \Gamma_t(x_t)} \{F_t(x_t, x') + V_{t+1}(x')\}$$

- ▶ Let x_{t+1} be a solution to $\sup_{x' \in \Gamma_t(x_t)} \{F_t(x_t, x') + V_{t+1}(x')\}$ for all $t = 0, 1, \dots, T-1$. Using $V_T(x_T) = 0$, iterating on the Bellman equation gives

$$V_0(x_0) = F_0(x_0, x_1) + V_1(x_1) = F_0(x_0, x_1) + F_1(x_1, x_2) + V_2(x_2) = \dots = \sum_{t=0}^{T-1} F_t(x_t, x_{t+1})$$

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An alternative (more general) formulation

$$\begin{aligned} \sup_{\{u_t\}_{t=0}^{T-1}} \quad & \sum_{t=0}^{T-1} F_t(x_t, u_t) \\ \text{s.t.} \quad & u_t \in G_t(x_t) \\ & x_{t+1} = f_t(x_t, u_t) \\ & x_0 \text{ given} \end{aligned}$$

- ▶ $u_t \in U$ is the control variable
- ▶ $F_t : X \times U \rightarrow \mathbb{R}$ is the instantaneous payoff function
- ▶ $G_t(x)$ is the feasible correspondence: $G_t : X \rightrightarrows U$
- ▶ $f_t : X \times U \rightarrow X$ gives the next period state

The value function is

$$V_t(x) := \sup_{\{u_s\}_{s=t}^{T-1}} \sum_{s=t}^{T-1} F_s(x_s, u_s),$$

s.t. $u_s \in G_s(x_s)$ for all $s = t, \dots, T-1$

$x_{s+1} = f_s(x_s, u_s)$ for all $s = t, \dots, T-1$

$x_t = x$

Dynamic Programming Principle

If a control sequence $(u_0, u_1, \dots, u_{T-1})$ with induced states $(x_1, x_2, \dots, x_T)$ is a solution to problem $V_0(x_0)$, then for any $t = 1, 2, \dots, T - 1$, $(u_t, \dots, u_{T-1})$ is a solution to problem $V_t(x_t)$.

Bellman Equation

For all $t = 0, 1, \dots, T - 1$, we have

$$V_t(x) = \sup_{u \in G_t(x)} \{F_t(x, u) + V_{t+1}(f_t(x, u))\}$$

Principle of Optimality

The control sequence $(u_0, \dots, u_{T-1})$ with induced states $(x_1, x_2, \dots, x_T)$ is a solution to $V_0(x_0)$ if and only if for all $t = 0, 1, \dots, T - 1$, u_t is a solution to

$$\sup_{u \in G_t(x_t)} \{F_t(x_t, u) + V_{t+1}(f_t(x_t, u))\}$$

where $x_{t+1} = f_t(x_t, u_t)$.

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- ▶ Consider again the cake eating problem with CRRA utility and $T = 3$
- ▶ The Bellman equation is

$$V_t(x_t) = \max_{x_{t+1} \in [0, x_t]} \left\{ \beta^t \frac{(x_t - x_{t+1})^{1-\gamma}}{1-\gamma} + V_{t+1}(x_{t+1}) \right\}$$

- ▶ We use max here because the optimum exists (continuous function on compact set)
- ▶ Solve it via backward induction

- ▶ When $t = 2$:

$$V_2(x_2) = \max_{x_3 \in [0, x_2]} \beta^2 \frac{(x_2 - x_3)^{1-\gamma}}{1-\gamma} = \beta^2 \frac{x_2^{1-\gamma}}{1-\gamma}$$

and the optimal policy is $x_3 = 0$: eat everything remaining (no reason to save)

- ▶ When $t = 1$:

$$V_1(x_1) = \max_{x_2 \in [0, x_1]} \left\{ \beta \frac{(x_1 - x_2)^{1-\gamma}}{1-\gamma} + \beta^2 \frac{x_2^{1-\gamma}}{1-\gamma} \right\} = \frac{x_1^{1-\gamma}}{1-\gamma} \frac{\beta}{(1 + \beta^{\frac{1}{\gamma}})^{-\gamma}}$$

with $x_2 = x_1 / (1 + \beta^{-\frac{1}{\gamma}})$

- ▶ When $t = 0$:

$$V_0(x_0) = \max_{x_1 \in [0, x_0]} \left\{ \frac{(x_0 - x_1)^{1-\gamma}}{1-\gamma} + \frac{\beta}{(1 + \beta^{\frac{1}{\gamma}})^{-\gamma}} \frac{x_1^{1-\gamma}}{1-\gamma} \right\}$$

which can be solved although a bit tedious

- ▶ Let $\Gamma_0(x_0) := [0, x_0^4 + 2x_0 + 3]$, $\Gamma_1(x_1) := [\frac{x_1}{2}, x_1^2 + x_1]$, and $\Gamma_2(x_2) := [0, \frac{x_2^2 + 4}{x_2}]$
- ▶ Define $f(x_1, x_0) := 2x_1x_0 - x_1^2 + x_1$, $g(x_2, x_1) := -\frac{1}{2x_2} + x_2x_1 - \frac{x_2^2}{2}$, $h(x_3, x_2) := \sqrt{x_3} - \frac{x_3x_2}{2}$
- ▶ Consider the following problem

$$\begin{aligned} & \sup_{x_1, x_2, x_3} \{f(x_1, x_0) + g(x_2, x_1) + h(x_3, x_2)\} \\ & \text{s.t. } x_i \in \Gamma_{i-1}(x_{i-1}) \text{ for all } i = 1, 2, 3 \\ & \quad x_0 \geq 0 \text{ given} \end{aligned}$$

- ▶ This is a 3-stage problem with state-dependent constraint sets
- ▶ The payoff functions and feasibility constraints have different functional forms at each stage

Stage 3:

- Compute $V_2(x_2)$

$$V_2(x_2) = \sup_{y \in \Gamma_2(x_2)} h(y, x_2) = \sup_{y \in \Gamma_2(x_2)} \left\{ \sqrt{y} - \frac{yx_2}{2} \right\}$$

- First-order condition:

$$\frac{d}{dy} \left(\sqrt{y} - \frac{yx_2}{2} \right) = \frac{1}{2\sqrt{y}} - \frac{x_2}{2} = 0$$

- Solving:

$$\frac{1}{2\sqrt{y}} = \frac{x_2}{2} \implies y^* = \frac{1}{x_2^2}$$

- Check second order condition: $-\frac{1}{4y^{3/2}} < 0$, so this is indeed a maximum
- Since $y^* \in \Gamma_2(x_2)$, $V_2(x_2) = \frac{1}{2x_2}$

Stage 2:

- Compute $V_1(x_1)$:

$$\begin{aligned} V_1(x_1) &= \sup_{y \in \Gamma_1(x_1)} \{g(y, x_1) + V_2(y)\} \\ &= \sup_{y \in \Gamma_1(x_1)} \left\{ -\frac{1}{2y} + yx_1 - \frac{y^2}{2} + \frac{1}{2y} \right\} \end{aligned}$$

- First-order condition gives $y^* = x_1$
- Second-order condition is $-1 < 0$
- Check feasibility: $y^* = x_1$ must satisfy $\frac{x_1}{2} \leq x_1 \leq x_1^2 + x_1$, which holds for $x_1 \geq 0$
- The value function is $V_1(x_1) = x_1^2/2$

Stage 1:

- Compute $V_0(x_0)$:

$$V_0(x_0) = \sup_{y \in \Gamma_0(x_0)} \left\{ 2yx_0 + y - \frac{y^2}{2} \right\}$$

- First-order condition gives $y = 2x_0 + 1$
- Second-order condition: $-1 < 0$
- Check feasibility: $2x_0 + 1 \in [0, x_0^4 + 2x_0 + 3]$
- The value function is $V_0(x_0) = 2x_0^2 + 2x_0 + 1/2$

The optimal policy is:

$$\left(2x_0 + 1, 2x_0 + 1, \frac{1}{(2x_0 + 1)^2} \right)$$